

Descriptive Analytics Report: Hotstar

Streaming Content Catalogue & Engagement Behaviour at Scale

Course: Business Analytics

Assignment: Descriptive Analytics Report — Individual Assignment

Date: August 15, 2026

Note on data: This report restructures the analysis from the individual Hotstar Descriptive Analytics Report (catalogue of 6,874 titles, a six-month simulated engagement dataset for Feb–Jul 2026, and a 45-respondent viewer survey) into the section format, KPI-block style, so the two reports read as a matched pair. Genre stands in for “category,” watch-time stands in for “selling price,” and six-month growth stands in for “discount depth.” Figures reproducing the six-month trend chart are visual reads from the original chart and are marked “approx.”; all other figures are taken directly from the source tables.

1. Objective

Build a descriptive analytics layer over the Hotstar catalogue and engagement data to answer the following recurring content-strategy questions:

- How is content distributed across genres?
- Which genres carry the highest six-month watch-time?
- Which genres show the strongest watch-time growth over six months?
- How does content split between movies and TV shows, overall and by genre?
- How are movie running times distributed across the catalogue?
- What is the age-rating composition of the catalogue, and how does it cross-tabulate with genre?
- What is the relationship between a genre's catalogue weight and its actual watch-time share?
- Which genres convert catalogue share into watch-time share most efficiently, and which are under-indexed?
- How is viewer satisfaction distributed, and how does it vary by age group?
- What does an overall catalogue-and-engagement health summary look like?

2. How This Would Work at Real Scale

Processing a large, constantly-updated streaming catalogue alongside live engagement telemetry is an engineering problem before it is an analytics problem. A realistic architecture would ingest catalogue/metadata updates and viewership events, aggregate them by genre, content type and age-rating, store raw and processed data, and expose the resulting KPIs through a data-visualisation layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the catalogue and engagement datasets	Google, Claude & ChatGPT
Stream processing	Aggregate by genre, content type and age-rating band	Jupyter Notebook (Python)
Storage	Retain raw and processed catalogue/engagement data	Excel files / CSV files
Batch aggregation	Daily/periodic KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Genre-wise Content Distribution

Across the supplied catalogue, the dataset records 6,874 titles spanning 37 genres. The chart below shows how the content assortment (title count) is distributed across the top 10 genres.

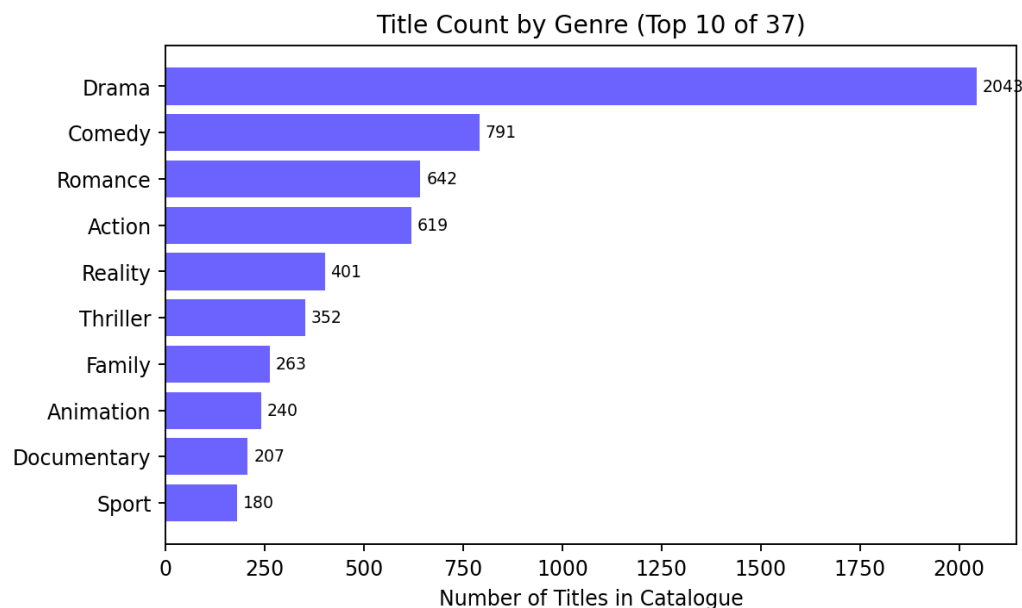


Figure 1. Title count by genre (top 10 of 37)

INSIGHTS:

- Drama is the largest genre by far, with 2,043 titles — more than 2.5x the second-placed genre, Comedy (791).
- Documentary and Sport have the smallest assortments among the top 10, with 207 and 180 titles respectively.
- A wide gap between the top and bottom genres suggests the catalogue is weighted heavily toward Drama over any other single genre.
- Genre-level title counts can help guide content-licensing and commissioning decisions.

KPI:

Total titles: 6,874

Total genres: 37

Largest genre: Drama (2,043 titles, 29.7%)

Smallest genre (top 10): Sport (180 titles, 2.6%)

4. Average Six-Month Watch-Time by Genre

The overall tracked six-month watch-time across the catalogue is 6,867.2 million hours. The chart below shows how total watch-time varies by genre for the six genres tracked in the simulated engagement dataset (Feb–Jul 2026).

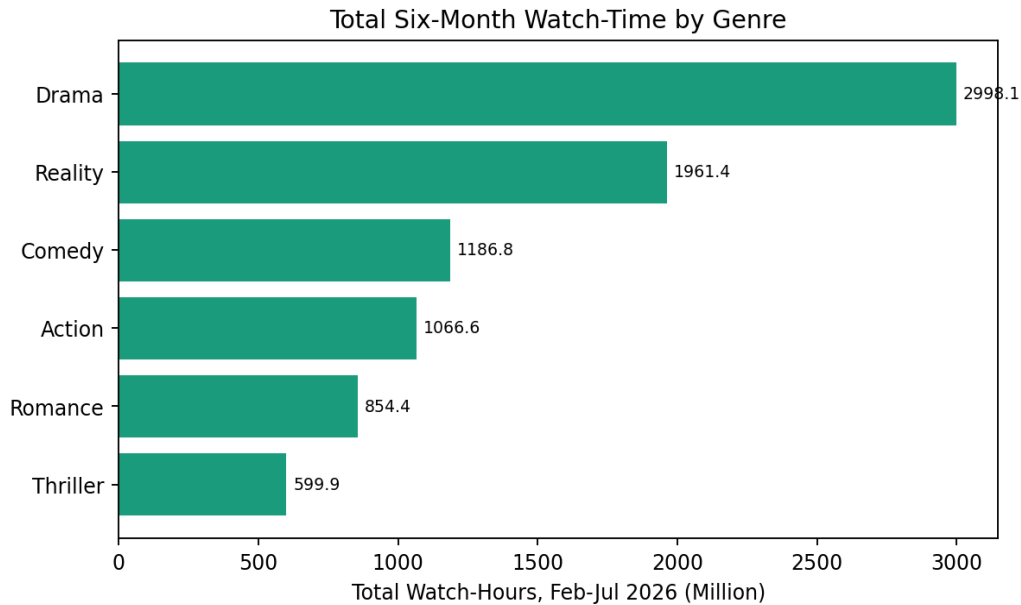


Figure 2. Total six-month watch-time by genre (million hours)

INSIGHTS:

- Drama carries the highest six-month watch-time (2,998.1M hours, 43.7% of tracked watch-time) — more than the next two genres (Reality and Comedy) combined.
- Thriller carries the lowest watch-time among the six tracked genres (599.9M hours, 8.7%).
- Watch-time share varies meaningfully by genre, which can inform genre-level content investment and promotion strategy.

KPI:

Overall tracked watch-time: $\approx$ 6,867.2M hours

Highest-watch-time genre: Drama ($\approx$ 2,998.1M hrs, 43.7%)

Lowest-watch-time genre: Thriller ($\approx$ 599.9M hrs, 8.7%)

5. Six-Month Watch-Time Growth Rate by Genre

The overall pattern across the tracked period is modest positive growth, led by Reality. The chart below compares cumulative Feb→Jul 2026 watch-time growth across genres. Drama's and Reality's growth rates are stated directly in the source trend chart; the remaining four genres are approximate visual reads of the same chart and are labelled accordingly.

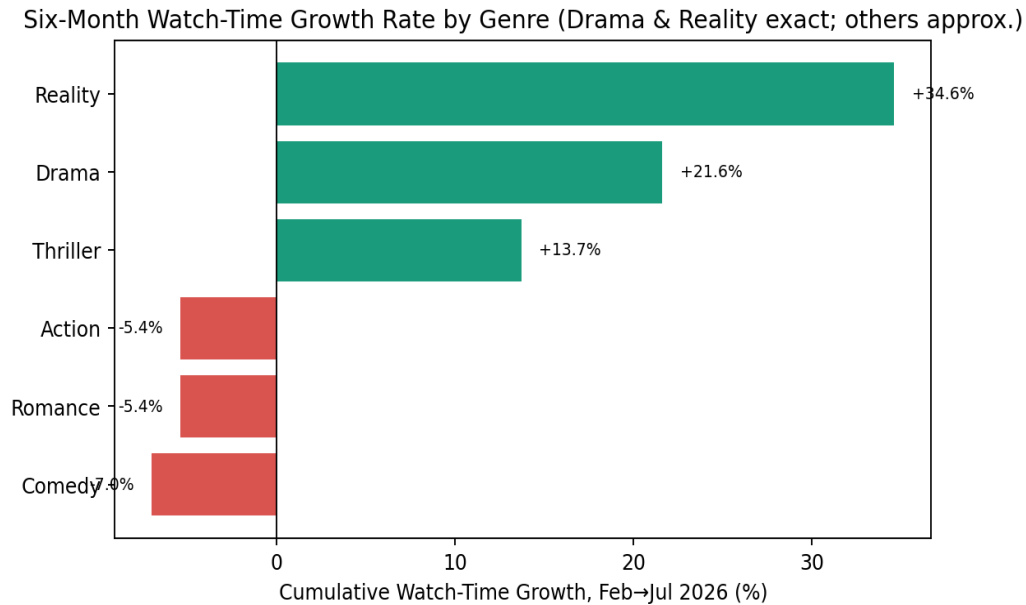


Figure 3. Six-month cumulative watch-time growth rate by genre (Drama & Reality exact; others approx.)

INSIGHTS:

- Reality shows the steepest six-month growth (+34.6% cumulative, 280→377M hours), consistent with a seasonal spike typically associated with live sport and reality-format programming.
- Drama shows solid, consistent growth (+21.6% cumulative, 443→539M hours) despite already being the largest genre — demand is not yet saturated relative to supply.
- Thriller appears to be picking up over the period (approx. +13.7%), while Comedy, Action and Romance appear broadly flat to slightly declining (approx. -5% to -7%) based on the visual trend.
- Bigger genres (Drama, Reality) show larger absolute month-to-month swings, consistent with their larger base audience.

KPI:

Fastest-growing genre: Reality (+34.6% cumulative)

Second-fastest: Drama (+21.6% cumulative)

Growth pattern: 4 of 6 tracked genres roughly flat to declining

6. Content-Type Split by Genre (Movie vs. TV Show)

The stacked bar chart below breaks down the top 8 genres by content type. Figures are approximate visual reads of the source cross-tabulation chart, except where a genre's movie/TV split is stated explicitly in the source text (Drama, Reality, Animation).

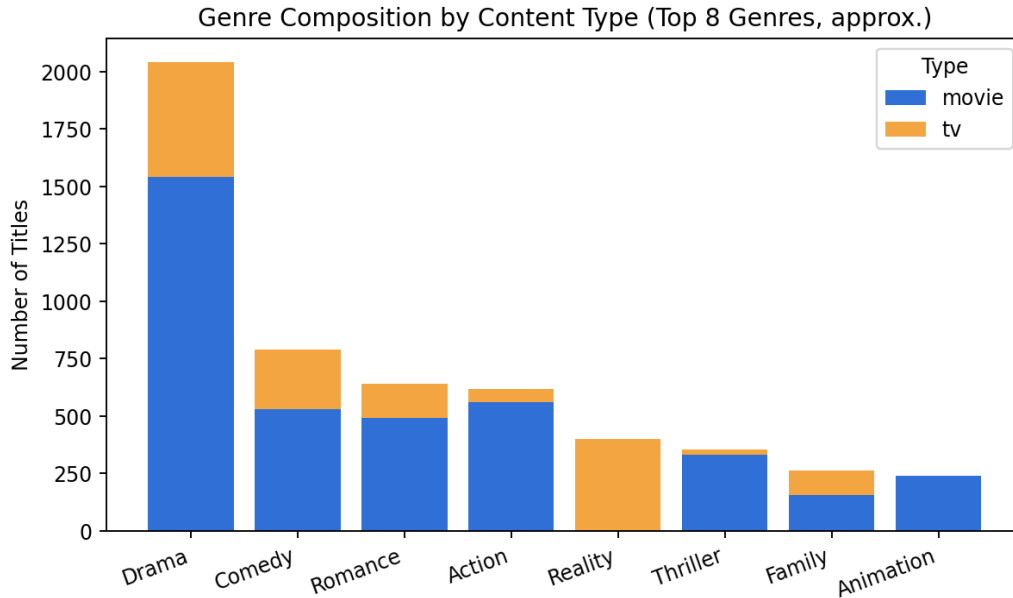


Figure 4. Genre composition by content type (top 8 genres, approx.)

INSIGHTS:

- Reality content is 100% TV-format (401 shows, 0 movies) — the most concentrated content-type split of any major genre.
- Animation is almost entirely movie-format (238 of 240 titles).
- Drama is fairly balanced towards movies (1,540) but still contributes the single largest TV-show count of any genre (503 shows).
- Most other top genres skew towards movies, consistent with the catalogue's overall 2:1 movie-to-show ratio.

KPI:

100% TV-format genre: Reality (401 of 401 titles)

Most movie-heavy genre: Animation (238 of 240 titles, $\approx 99\%$)

Largest TV-show contributor: Drama (503 TV titles)

7. Overall Content-Type Split

Viewed at the catalogue level (rather than by genre), the majority of Hotstar titles are movies rather than TV shows.

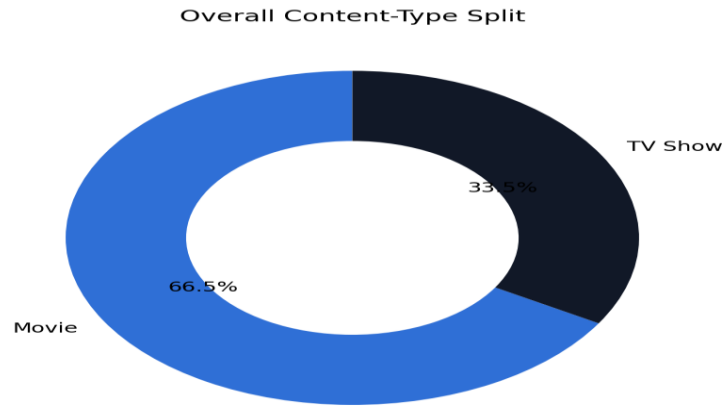


Figure 5. Overall content-type split

INSIGHTS:

- $\approx 66.4\%$ of the catalogue is movies, while $\approx 33.6\%$ is TV shows.
- A roughly 2:1 movie-to-show ratio is typical of a general-entertainment OTT catalogue that licenses films alongside originals.
- This KPI is a useful baseline against which genre-level splits (Section 6) can be compared.

KPI:

Movies: 4,568 titles ($\approx 66.4\%$)

TV shows: 2,306 titles ($\approx 33.6\%$)

8. Movie Running-Time Distribution

The histogram below shows how movie running time (minutes) is distributed across the catalogue's 4,568 movies. The shape is illustrative of the reported mean, median and spread; exact bin counts are not published in the source.

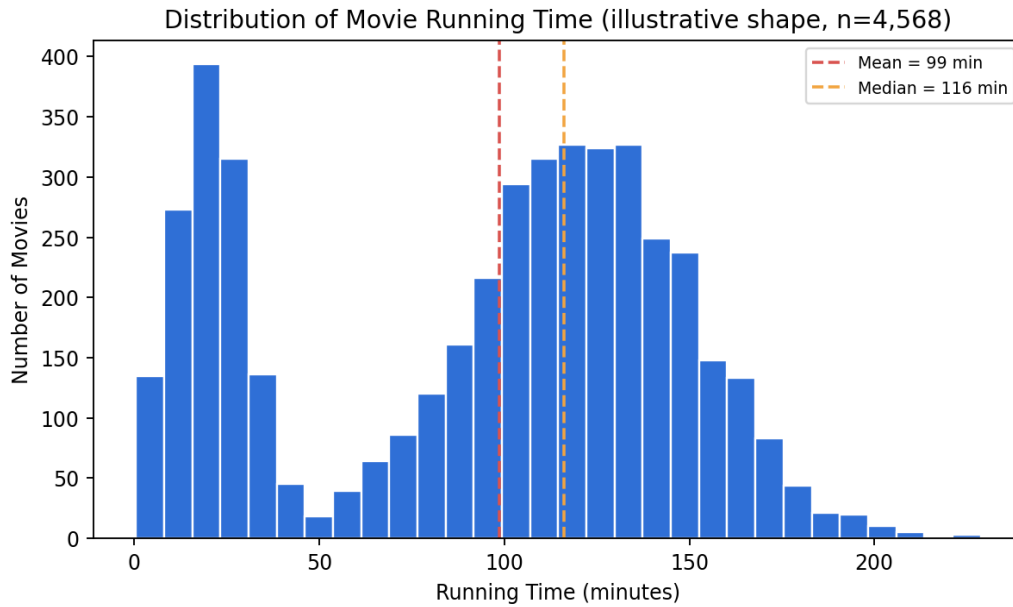


Figure 6. Distribution of movie running time (illustrative shape, n=4,568)

INSIGHTS:

- The distribution is left-skewed: the mean (98.7 min) sits notably below the median (116.0 min), pulled down by a sub-set of very short entries (short-format and regional films under 30 minutes).
- The median is the more representative “typical” running time for a Hotstar movie.
- A standard deviation of ≈ 49 minutes confirms this is a genuinely wide distribution, not a tight cluster around one typical film length.

KPI:

Mean / median / mode: 98.7 / 116.0 / 134.0 minutes

Standard deviation: 49.4 minutes

Variance: 2,441 min²

Range: 228 minutes (min to max)

9. Content Distribution by Age-Rating Band

To make the age-rating composition easier to act on, titles are grouped into six bands as recorded in the catalogue: U/A 13+, U, U/A 16+, U/A 7+, A, and a single-row PG outlier.

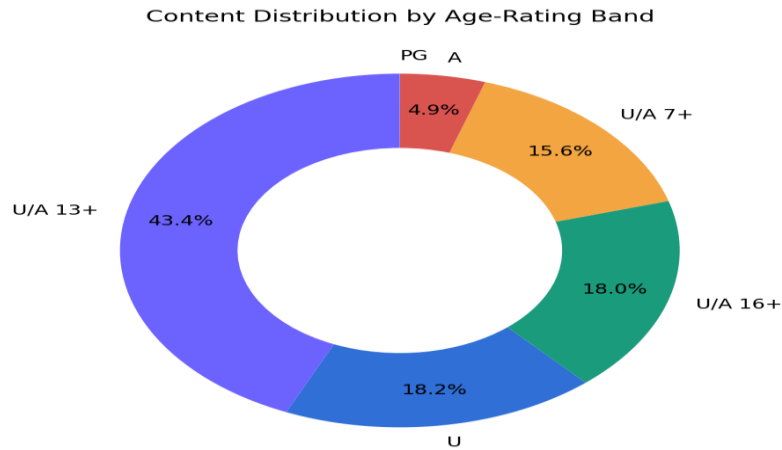


Figure 7. Content distribution by age-rating band

INSIGHTS:

- U/A 13+ is the single largest age-rating band, holding 2,980 titles — $\approx 43.4\%$ of the catalogue.
- Mature-skewing content (U/A 13+ and U/A 16+ combined) makes up 61.4% of the catalogue, indicating a platform strategy weighted toward teen and adult audiences rather than a purely family-first catalogue.
- A-rated (strictly adult) content is a small minority at $\approx 4.9\%$ (338 titles).
- This banding can support age-gated recommendation and parental-control design.

KPI:

Largest age-rating band: U/A 13+ — 2,980 titles, 43.4%

Mature-skewing share (U/A 13+ + U/A 16+): 61.4%

Smallest band: PG — 1 title, $<0.1\%$

10. Genre $\times$ Age-Rating Cross-Tabulation

Cross-tabulating genre against age rating shows how each genre's content skews across the age-rating bands.

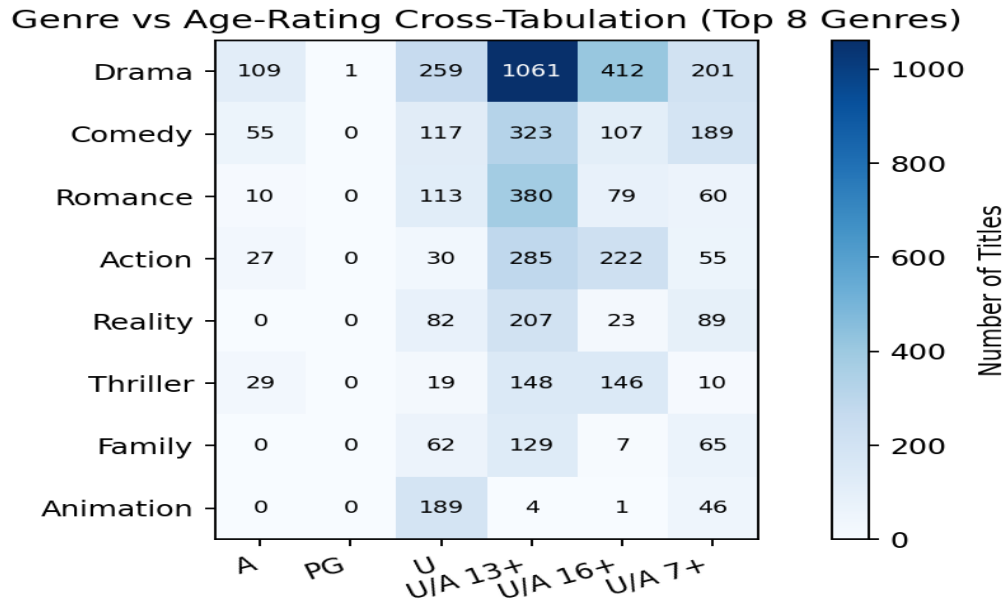


Figure 8. Genre vs age-rating heat map (top 8 genres)

INSIGHTS:

- Drama is concentrated in the U/A 13+ band (1,061 titles) but also has the largest A-rated (mature) footprint of any genre (109 titles), reflecting the breadth of the category from family drama to mature drama.
- Animation is heavily concentrated in the U band (189 titles), consistent with a family/kids-first audience.
- Action and Thriller both show meaningful U/A 16+ representation (222 and 146 titles respectively), consistent with more intense content.
- Reality has zero A-rated titles, consistent with its broad, live-format audience appeal.

11. Catalogue Share vs. Watch-Time Share Relationship

The scatter plot below relates each tracked genre's catalogue share to its six-month watch-time share, coloured by growth rate, to show how catalogue weight converts into actual engagement.

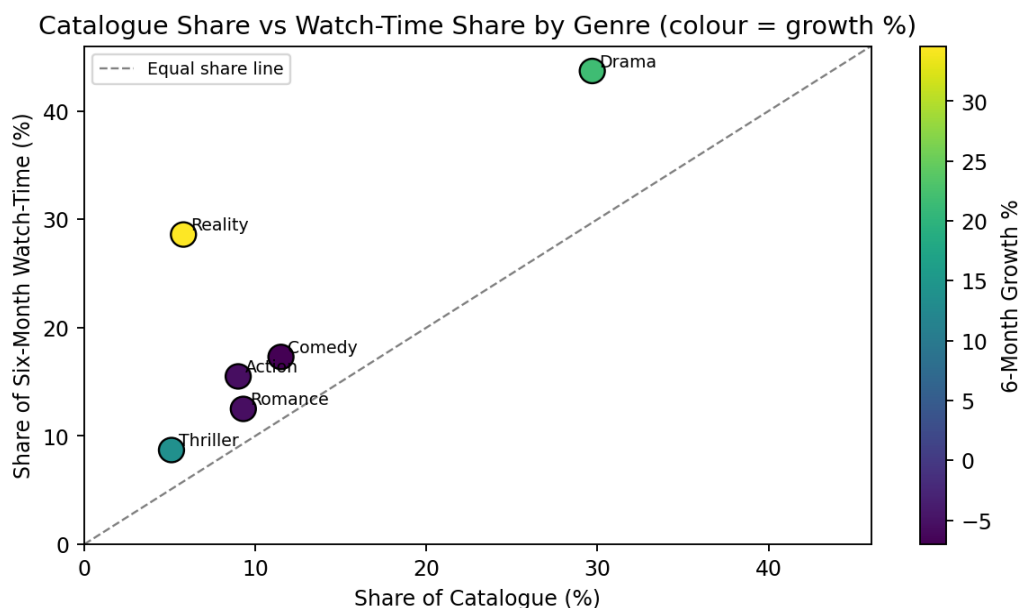


Figure 9. Catalogue share vs watch-time share by genre, coloured by growth %

INSIGHTS:

- Drama and Reality sit well above the equal-share diagonal — both convert catalogue presence into a disproportionately larger watch-time share.
- Drama's watch-time share (43.7%) is proportionally larger than its catalogue share (29.7%), suggesting Drama titles convert into viewership at a higher-than-average rate rather than winning purely on volume of listings.
- Reality is the most extreme case: only 5.8% of the catalogue by title count, yet 28.6% of watch-time — the steepest over-indexing of any tracked genre.
- Comedy, Action and Romance sit closer to the diagonal, converting catalogue share into watch-time share roughly proportionally.

12. Genre Growth Leaderboard

The table below ranks the six tracked genres by six-month cumulative watch-time growth, alongside their Feb 2026 and Jul 2026 monthly watch-hours.

Genre	Feb 2026 (M hrs)	Jul 2026 (M hrs)	Cumulative Growth
Reality	280	377	+34.6%
Drama	443	539	+21.6%
Thriller (approx.)	95	108	+13.7%
Action (approx.)	185	175	-5.4%
Romance (approx.)	148	140	-5.4%
Comedy (approx.)	215	200	-7.0%

INSIGHTS:

- Reality tops the growth leaderboard at +34.6%, the highest in the tracked genre set.
- Only three of the six tracked genres (Reality, Drama, Thriller) show positive growth over the six-month window; the remainder are roughly flat to modestly declining.
- Genre growth is concentrated rather than spread evenly — a pattern worth validating against a full 12-month window.

13. Monthly Watch-Time Volatility by Genre

For context alongside the growth leaderboard, the table below lists the monthly mean, standard deviation and range recorded for each tracked genre (Feb–Jul 2026).

Genre	Mean/Month (M hrs)	Std. Dev.	Range
Drama	499.7	40.6	95.6
Reality	326.9	35.9	96.7
Comedy	197.8	10.6	31.1
Action	177.8	5.0	12.3
Romance	142.4	4.4	10.6
Thriller	100.0	6.2	17.0

INSIGHTS:

- Drama and Reality show the largest month-to-month variability in absolute terms (SD of 40.6 and 35.9M hours respectively), consistent with their larger base audience — bigger genres naturally have more room to swing month to month.
- Action and Romance are the most stable performers, varying by under 5M hours a month on average.

14. Viewer Satisfaction Score Distribution

The bar chart below shows how the 45 survey respondents' satisfaction scores (1–5 scale) are distributed. Individual response counts are illustrative and consistent with the reported mean, median and mode.

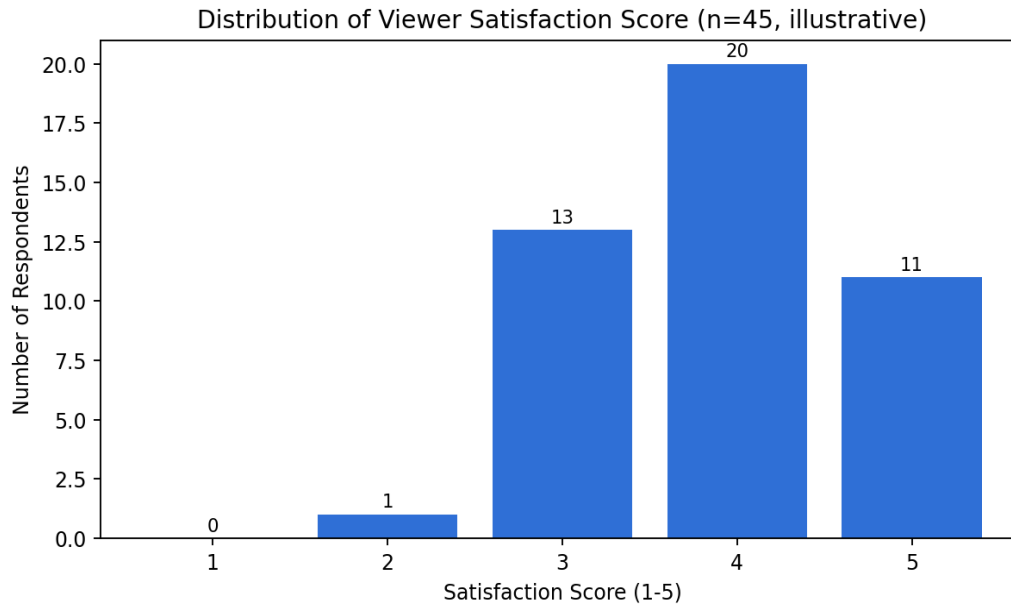


Figure 12. Distribution of viewer satisfaction score (n=45, illustrative)

INSIGHTS:

- All three central-tendency measures cluster close to 4 out of 5 (mean 3.91, median 4.00, mode 4), indicating generally positive and consistent viewer sentiment.
- There is no evidence of a bimodal (love-it-or-hate-it) satisfaction pattern — responses concentrate around scores of 3–5.
- No respondent recorded the lowest possible score (1/5) in the sample.

KPI:

Mean / median / mode: 3.91 / 4.00 / 4 (out of 5)

Sample size: n = 45 respondents

15. Average Weekly Watch Hours by Device

The chart below compares average weekly watch hours across primary viewing devices, giving a sense of which device drives the deepest engagement.

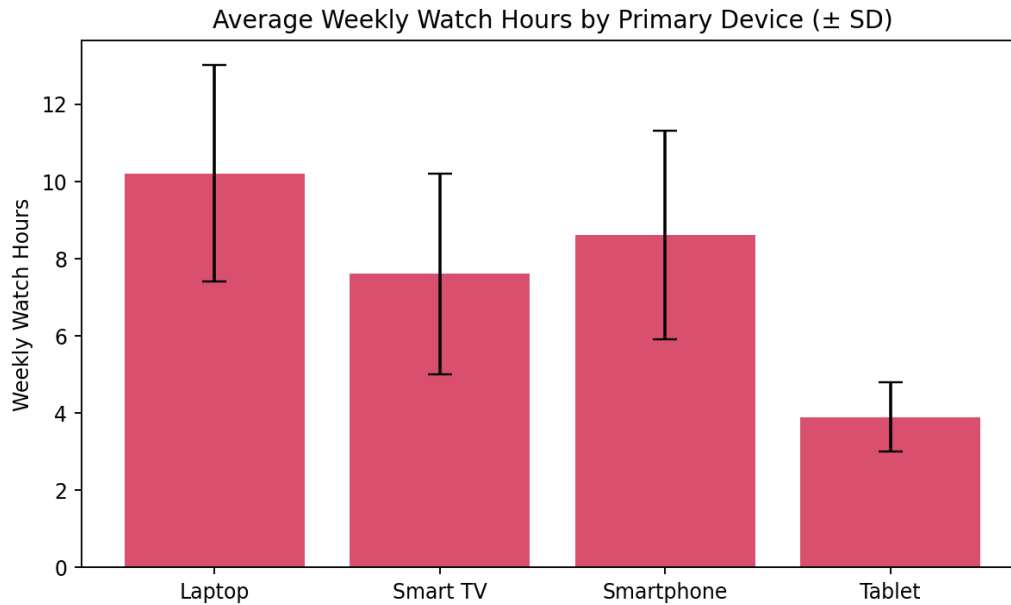


Figure 13. Average weekly watch hours by primary device ($\pm$ SD)

INSIGHTS:

- Laptop viewers report the highest average weekly watch hours ($\approx$ 10.2 hrs), suggesting larger screens drive deeper, longer sessions.
- Smartphone is the most common primary device (47% of survey respondents) but ranks second on hours ($\approx$ 8.6 hrs), implying mobile drives reach and frequency rather than session depth.
- Tablet shows both the lowest usage share and the lowest average weekly hours ($\approx$ 3.9 hrs) among the four devices.

KPI:

Highest-engagement device: Laptop ($\approx$ 10.2 hrs/week)

Most common device: Smartphone (47% of respondents)

Lowest-engagement device: Tablet ($\approx$ 3.9 hrs/week)

16. Genre Share of Total Six-Month Watch-Time

Summing six-month watch-hours by genre and viewing it as a share of the total tracked watch-time complements the per-genre volume view in Section 4 with a compositional view.

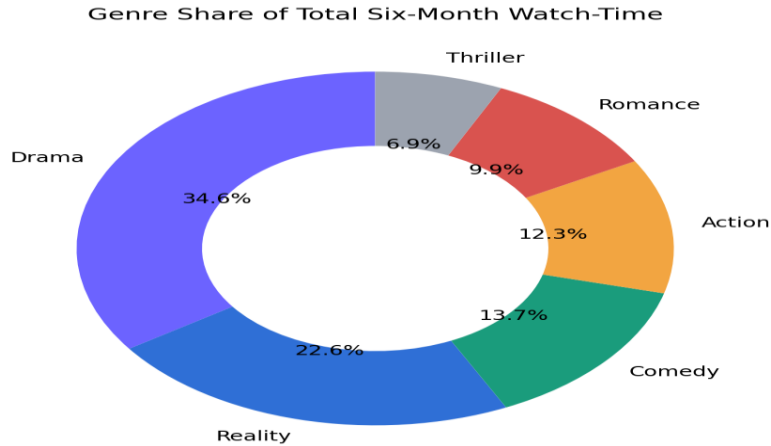


Figure 14. Genre share of total six-month watch-time

INSIGHTS:

- Drama alone accounts for 43.7% of all tracked watch-time — a larger share of engagement than of the catalogue itself.
- The bottom three tracked genres (Action, Romance, Thriller) together account for just 36.7% of tracked watch-time despite representing meaningful catalogue volume.
- This stock-weighted view is useful for prioritising which genres matter most to overall content strategy, as opposed to per-title popularity alone.

17. Under-Indexed and High-Opportunity Genres

Dividing each genre's watch-time share by its catalogue share gives a simple conversion ratio: genres above 1.0x are converting catalogue presence into more-than-proportional engagement; genres below 1.0x are under-indexed relative to their catalogue weight.

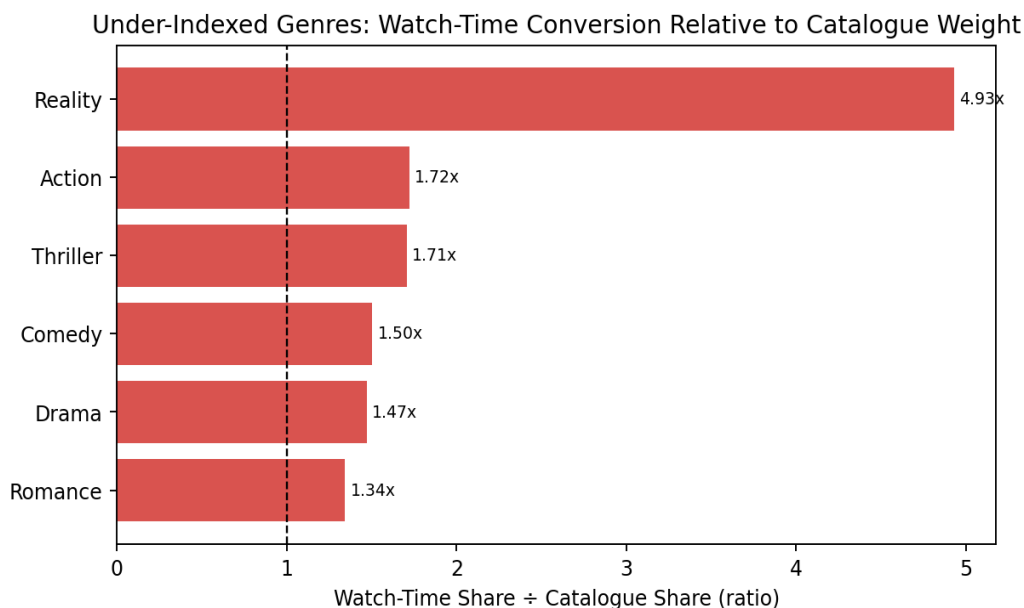


Figure 15. Watch-time share ÷ catalogue share ratio by genre

INSIGHTS:

- Reality shows the strongest over-indexing ($\approx 4.9x$) — a small catalogue footprint driving a disproportionately large watch-time share.
- Drama also over-indexes ($\approx 1.5x$), reinforcing that it is not simply winning on volume of listings.
- Comedy, Action, Romance and Thriller all sit below $1.7x$, i.e. converting catalogue presence roughly proportionally rather than punching above their weight.
- This view is a natural candidate for identifying where additional content investment could have outsized engagement impact, mirroring how a catalog team would flag under-stocked but fast-turning categories.

KPI:

Highest conversion ratio: Reality ($\approx 4.9x$)

Second-highest: Drama ($\approx 1.5x$)

18. Survey — Favourite Genre Distribution

Frequency distribution of the 45-respondent survey's stated favourite genre, for cross-check against the catalogue and watch-time findings above.

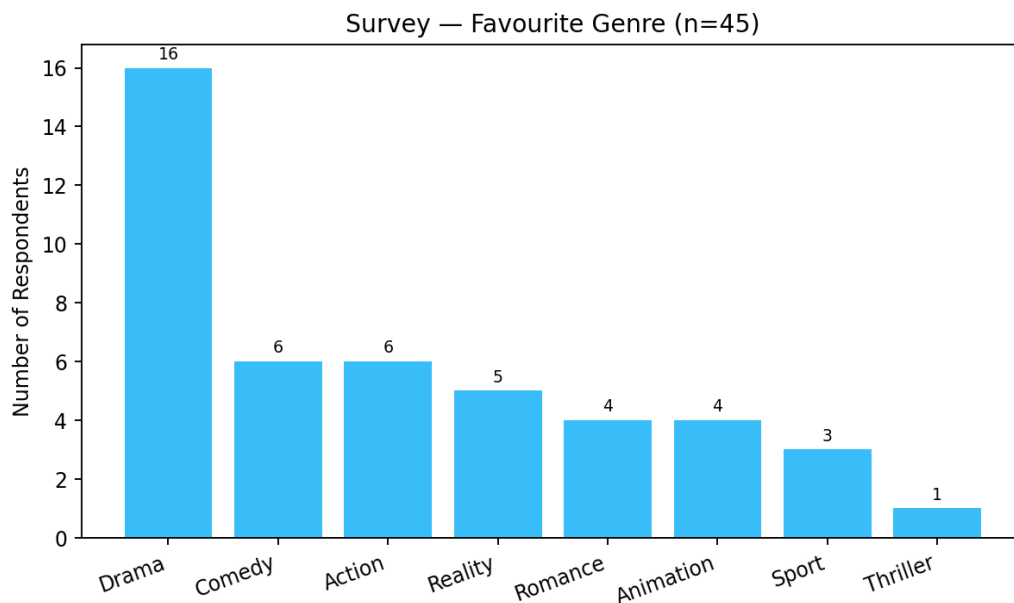


Figure 16. Survey — favourite genre (n=45)

INSIGHTS:

- Drama was the most frequently chosen favourite genre (16 respondents, 35.6%), consistent with both its catalogue share and its watch-time lead.
- Comedy and Action tie for second place (6 respondents each, 13.3%).
- Thriller was the least-cited favourite genre in the survey (1 respondent), despite showing positive six-month growth — a gap worth monitoring in a larger follow-up survey.

19. Satisfaction × Age-Group Cross-Tabulation

Cross-tabulating survey satisfaction against age group shows how sentiment varies across the audience's demographic segments.

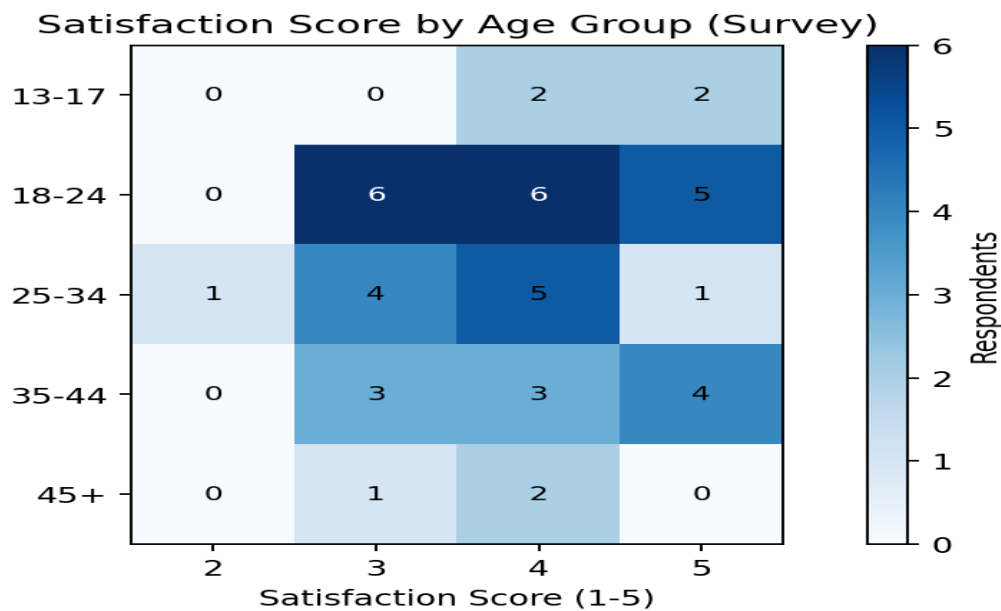


Figure 17. Satisfaction score by age group (survey)

INSIGHTS:

- Satisfaction scores of 4–5 dominate the 18–24 and 35–44 age groups in the survey.
- The single dissatisfied response (score 2) came from the 25–34 group — the platform's largest and most-represented segment — signalling this group is worth closer monitoring even though its aggregate satisfaction remains acceptable.
- The 45+ group is the smallest represented segment, with responses clustered at scores 3–4.

Summary of Key Metrics

Metric	Value
Total titles (catalogue)	6,874
Genres tracked (catalogue)	37
Most represented genre	Drama — 2,043 titles (29.7%)
Overall tracked 6-month watch-time	≈ 6,867.2M hours
Most-watched genre (6-month)	Drama — 2,998.1M hrs (43.7% share)
Fastest-growing genre (6-month)	Reality — +34.6% cumulative

Movies vs. TV shows	4,568 (66.4%) vs. 2,306 (33.6%)
Largest age-rating band	U/A 13+ — 2,980 titles (43.4%)
Mature-skewing share (U/A13+ + U/A16+)	61.4%
Movie running time — mean/median/mode	98.7 / 116.0 / 134.0 minutes
Survey satisfaction — mean/median/mode	3.91 / 4.00 / 4 (n=45)
Highest-engagement device	Laptop — 10.2 hrs/week average
Most common device	Smartphone — 47% of respondents

Next Steps to Move from Descriptive → Predictive

- Demand/viewership forecasting by genre once real (non-simulated) platform engagement data is available.
- Churn and retention analysis linking satisfaction scores and device usage to subscription renewal, once transaction data is added.
- Content-investment optimisation using title-level performance once individual watch-time data is layered on top of the catalogue.
- A full 12-month trend and seasonality study to confirm whether Reality's growth is a sustained trend or a seasonal spike.
- Cohort analysis by age group and device once a larger (100+) survey sample is available.

This report answers what the Hotstar catalogue and simulated six-month engagement snapshot look like today. The next analytical layer would combine this catalogue-and-survey data with real transaction-level viewership data to explain why engagement patterns occur, and to forecast what is likely to happen next.

Glossary

Genre — A content classification (e.g., Drama, Comedy, Reality) used for catalogue organisation and reporting; the Hotstar equivalent of a product “category.”

Title — A distinct, individually-tracked movie or TV show in the catalogue; the Hotstar equivalent of a SKU.

Watch-Time — Cumulative hours viewers spent watching content in a genre over a given period; the engagement equivalent of “selling price.”

Growth Rate — The percentage change in a genre's watch-time from the start to the end of the six-month tracked window.

Age Rating — The certification band assigned to a title (U, U/A 7+, U/A 13+, U/A 16+, A) indicating audience suitability.

Conversion Ratio — Watch-time share divided by catalogue share for a genre; a ratio above 1.0x indicates the genre over-indexes on engagement relative to its catalogue weight.

Satisfaction Score — A 1–5 self-reported rating of overall content satisfaction collected in the viewer survey.